

Interconnect Parasitics

This chapter provides an overview of various techniques for handling and representing interconnect parasitics for timing verification of the designs. In digital designs, a wire connecting pins of standard cells and blocks is referred to as a net. A net typically has only one driver while it can drive a number of fanout cells or blocks. After physical implementation, the net can travel on multiple metal layers of the chip. Various metal layers can have different resistance and capacitance values. For equivalent electrical representation, a net is typically broken up into segments with each segment represented by equivalent parasitics. We refer to an interconnect trace as a synonym to a segment, that is, it is part of a net on a specific metal layer.

4.1 RLC for Interconnect

The interconnect resistance comes from the interconnect traces in various metal layers and vias in the design implementation. Figure 4-1 shows example nets traversing various metal layers and vias. Thus, the interconnect resistance can be considered as resistance between the output pin of a cell and the input pins of the fanout cells.

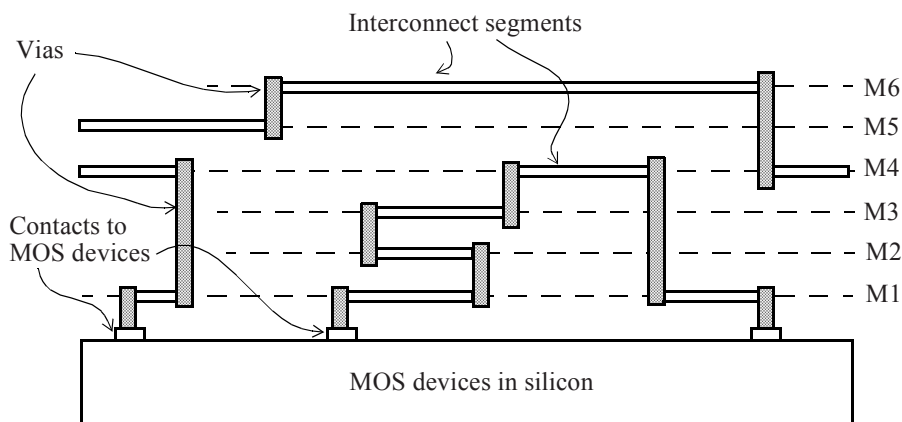


Figure 4-1 *Nets on metal layers.*

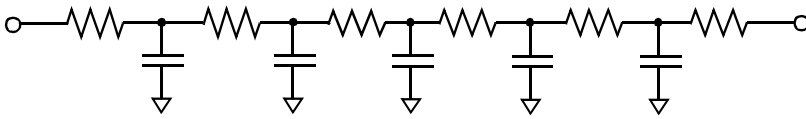
The interconnect capacitance contribution is also from the metal traces and is comprised of grounded capacitance as well as capacitance between neighboring signal routes.

The inductance arises due to *current* loops. Typically the effect of inductance can be ignored within the chip and is only considered for package and board level analysis. In chip level designs, the *current* loops are narrow and short - which means that the *current* return path is through a power or ground signal routed in close proximity. In most cases, the on-chip inductance is not considered for the timing analysis. Any further description of on-chip inductance analysis is beyond the scope of this book. Representation of interconnect resistance and capacitance is described next.

The resistance and capacitance (**RC**) for a section of interconnect trace is ideally represented by a distributed RC tree as shown in Figure 4-2. In this figure, the total resistance and capacitance of the RC tree - R_t and C_t respectively - correspond to $R_p * L$ and $C_p * L$ where R_p , C_p are per unit length values of interconnect resistance and capacitance for the trace and L is the trace length. The R_p , C_p values are typically obtained from the extracted parasitics for various configurations and is provided by the ASIC foundry.



(a) Trace of length L .



(b) Distributed RC tree.

Figure 4-2 *Interconnect trace.*

$$R_t = R_p * L$$

$$C_t = C_p * L$$

The RC interconnect can be represented by various simplified models. These are described in the subsections below.

T-model

In the T-model representation, the total capacitance C_t is modeled as connected halfway in the resistive tree. The total resistance R_t is broken in two sections (each being $R_t / 2$), with the C_t connection represented at the midpoint of the resistive tree as shown in Figure 4-3.

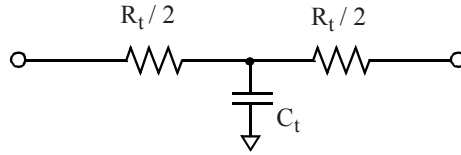


Figure 4-3 *T-model representation.*

Pi-model

In the Pi-model shown in Figure 4-4, the total capacitance C_t is broken into two sections (each being $C_t/2$) and connected on either side of the resistance.

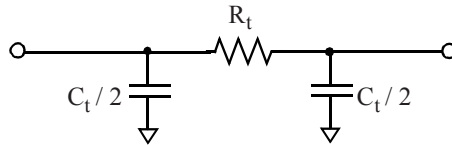


Figure 4-4 *Pi-model representation.*

More accurate representations of the distributed RC tree are obtained by breaking R_t and C_t into multiple sections. For N -sections, each of the intermediate sections of R and C are R_t/N and C_t/N . The end sections can be modeled along the concept of the T-model or the Pi-model. Figure 4-5 shows such an N -section with the end sections modeled using a T-model, while Figure 4-6 shows an N -section with the end sections modeled using the Pi-model.

With the broad overview of modeling of RC interconnect, we now describe how the parasitic interconnects are utilized during the pre-layout phase (by estimation) or post-layout phase (by detailed extraction). The next section describes the modeling of parasitic interconnect during the pre-layout process.

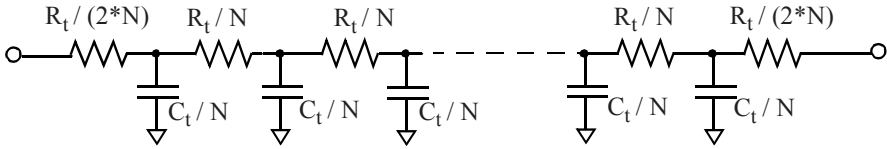


Figure 4-5 $R_t / (2 * N)$ on ends plus $(N-1)$ sections of R_t / N .

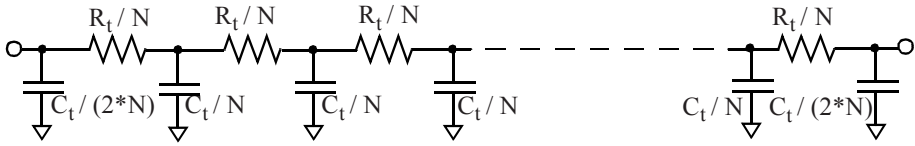


Figure 4-6 $C_t / (2 * N)$ on ends plus $(N-1)$ sections of C_t / N .

4.2 Wireload Models

Prior to floorplanning or layout, wireload models can be used to estimate capacitance, resistance and the area overhead due to interconnect. The wireload model is used to estimate the length of a net based upon the number of its fanouts. The wireload model depends upon the area of the block, and designs with different areas may choose different wireload models. The wireload model also maps the estimated length of the net into the resistance, capacitance and the corresponding area overhead due to routing.

The average wire length within a block correlates well with the size of the block; average net length increases as the block size is increased. Figure 4-7 shows that for different areas (chip or block size), different wireload models would typically be used in determining the parasitics. Thus, the figure depicts a smaller capacitance for the smaller sized block.

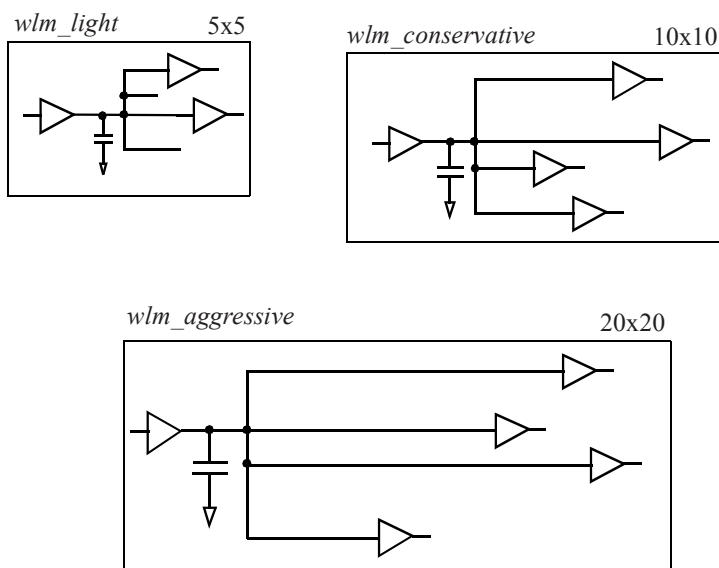


Figure 4-7 *Different wireload models for different areas.*

Here is an example of a wireload model.

```

wire_load ("wlm_conservative") {
  resistance : 5.0;
  capacitance : 1.1;
  area : 0.05;
  slope : 0.5;
  fanout_length (1, 2.6);
  fanout_length (2, 2.9);
  fanout_length (3, 3.2);
  fanout_length (4, 3.6);
  fanout_length (5, 4.1);
}

```

Resistance is resistance per unit length of the interconnect, *capacitance* is capacitance per unit length of the interconnect, and *area* is area overhead per

unit length of the interconnect. The *slope* is the extrapolation slope to be used for data points that are not specified in the fanout length table.

The wireload model illustrates how the length of the wire can be described as a function of fanout. The example above is depicted in Figure 4-8. For any fanout number not explicitly listed in the table, the interconnect length is obtained using linear extrapolation with the specified slope. For example, a fanout of 8 results in the following:

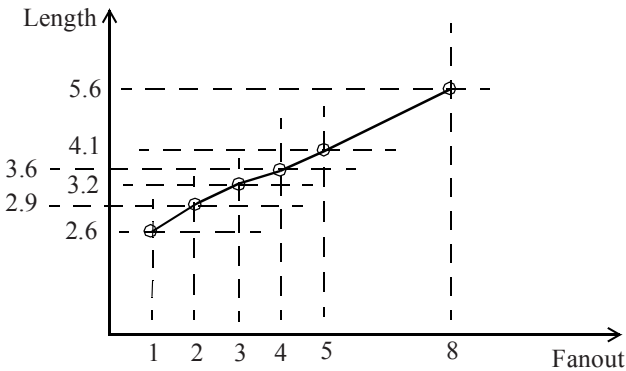


Figure 4-8 *Fanout vs wire length.*

$$\text{Length} = 4.1 + (8 - 5) * 0.5 = 5.6 \text{ units}$$

$$\text{Capacitance} = \text{Length} * \text{cap_coeff}(1.1) = 6.16 \text{ units}$$

$$\text{Resistance} = \text{Length} * \text{res_coeff}(5.0) = 28.0 \text{ units}$$

$$\begin{aligned} \text{Area overhead due to interconnect} &= \text{Length} * \text{area_coeff}(0.05) \\ &= 0.28 \text{ area units} \end{aligned}$$

The units for the length, capacitance, resistance and area are as specified in the library.

4.2.1 Interconnect Trees

Once the resistance and capacitance estimates, say R_{wire} and C_{wire} , of the pre-layout interconnect are determined, the next question is on the structure of the interconnect. How is the interconnect RC structure located with respect to the driving cell? This is important since the interconnect delay from a driver pin to a load pin depends upon how the interconnect is structured. In general, the interconnect delay depends upon the interconnect resistance and capacitance along the path. Thus, this delay can be different depending on the topology assumed for the net.

For pre-layout estimation, the interconnect RC tree can be represented using one of the following three different representations (see Figure 4-9). Note that the total interconnect length (and thus the resistance and capacitance estimates) is the same in each of the three cases.

- *Best-case tree:*

In the best-case tree, it is assumed that the destination (load) pin is physically adjacent to the driver. Thus, none of the wire resistance is in the path to the destination pin. All of the wire capacitance and the pin capacitances from other fanout pins still act as load on the driver pin.

- *Balanced tree:*

In this scenario, it is assumed that each destination pin is on a separate portion of the interconnect wire. Each path to the destination sees an equal portion of the total wire resistance and capacitance.

- *Worst-case tree:*

In this scenario, it is assumed that all the destination pins are together at the far end of the wire. Thus each destination pin sees the total wire resistance and the total wire capacitance.

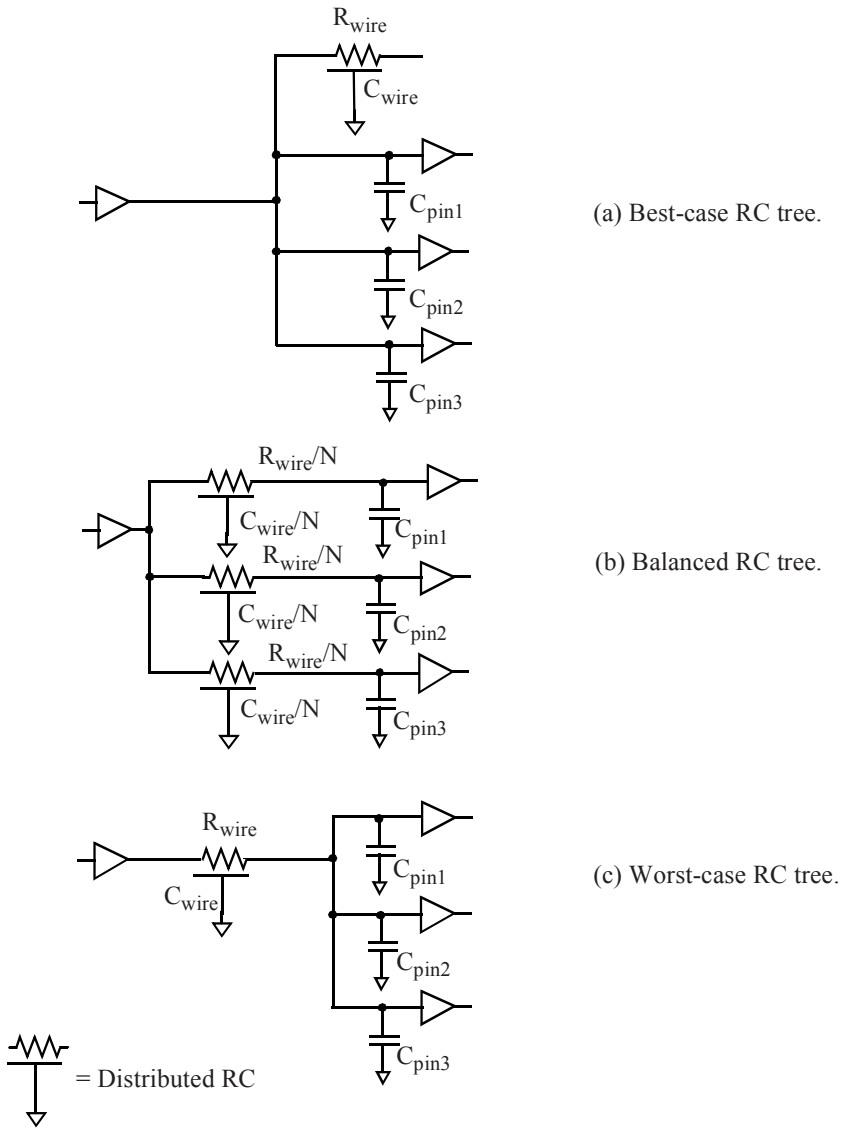


Figure 4-9 RC tree representations used during pre-layout.

4.2.2 Specifying Wireload Models

A wireload model is specified using the following command:

```
set_wire_load_model "wlm_cons" -library "lib_stdcell"  
# Says to use the wireload model wlm_cons present in the  
# cell library lib_stdcell.
```

When a net crosses a hierarchical boundary, different wireload models can be applied to different parts of the net in each hierarchical boundary based upon the **wireload mode**. These wireload modes are:

- i. top*
- ii. enclosed*
- iii. segmented*

The wireload mode can be specified using the *set_wire_load_mode* specification as shown below.

```
set_wire_load_mode enclosed
```

In the *top* wireload mode, all nets within the hierarchy inherit the wireload model of the top-level, that is, any wireload models specified in lower-level blocks are ignored. Thus, the top-level wireload model takes precedence. For the example shown in Figure 4-10, the *wlm_cons* wireload model specified in block *B1* takes precedence over all the other wireload models specified in blocks *B2*, *B3* and *B4*.

In the *enclosed* wireload mode, the wireload model of the block that fully encompasses the net is used for the entire net. For the example shown in Figure 4-11, the net *NETQ* is subsumed in block *B2* and thus the wireload model of block *B2*, *wlm_light*, is used for this net. Other nets which are fully contained in block *B3* use the *wlm_aggr* wireload model, whereas nets fully contained within block *B5* use the *wlm_typ* wireload model.

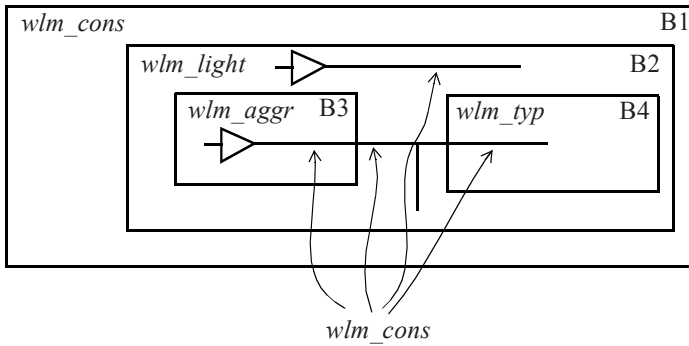


Figure 4-10 *TOP wireload mode.*

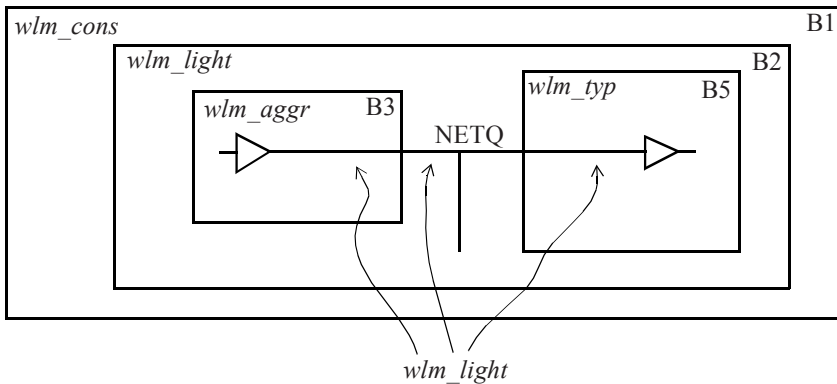


Figure 4-11 *ENCLOSED wireload mode.*

In the *segmented* wireload mode, each segment of the net gets its wireload model from the block that encompasses the net segment. Each portion of the net uses the appropriate wireload model within that level. Figure 4-12 illustrates an example of a net *NETQ* that has segments in three blocks. The interconnect for the fanout of this net within block B3 uses the wireload model *wlm_aggr*, the segment of net within block B4 utilizes the wireload

model *wlm_typ*, and the segment within block *B2* uses the wireload model *wlm_light*.

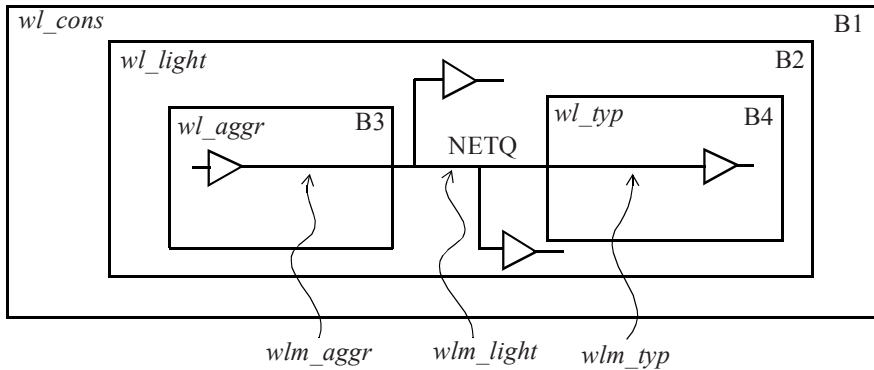


Figure 4-12 *SEGMENTED wireload mode.*

Typically a wireload model is selected based upon the chip area of the block. However these can be modified or changed at the user's discretion. For example, one can select the wireload model *wlm_aggr* for a block area between 0 and 400, the wireload model *wlm_typ* for area between 400 and 1000, and the wireload model *wlm_cons* for area 1000 or higher. Wireload models are typically defined in a cell library - however a user can define a custom wireload model as well. A **default wireload model** may optionally be specified in the cell library as:

```
default_wire_load: "wlm_light";
```

A **wireload selection group**, which selects a wireload model based upon area, is defined in a cell library. Here is one such example:

```
wire_load_selection (WireAreaSelGrp){
  wire_load_from_area(0, 50000, "wlm_light");
  wire_load_from_area(50000, 100000, "wlm_cons");
  wire_load_from_area(100000, 200000, "wlm_typ");
}
```

```
wire_load_from_area(200000, 500000, "wlm_aggr");  
}
```

A cell library can contain many such selection groups. A particular one can be selected for use in STA by using the *set_wire_load_selection_group* specification.

```
set_wire_load_selection_group WireAreaSelGrp
```

This section described the modeling of estimated parasitics before the physical implementation, that is, during the pre-layout phase. The next section describes the representation of parasitics extracted from the layout.

4.3 Representation of Extracted Parasitics

Parasitics extracted from a layout can be described in three formats:

- i. Detailed Standard Parasitic Format (DSPF)
- ii. Reduced Standard Parasitic Format (RSPF)
- iii. Standard Parasitic Extraction Format (SPEF)

Some tools provide a proprietary binary representation of the parasitics, such as SBPF; this helps in keeping the file size smaller and speeds up the reading of the parasitics by the tools. A brief description of each of the above three formats follows.

4.3.1 Detailed Standard Parasitic Format

In the DSPF representation, the detailed parasitics are represented in SPICE¹ format. The SPICE *Comment* statements are used to indicate the cell

1. Format that is readable by a circuit simulator, such as SPICE. Refer to [NAG75] or any book on analog integrated circuits design or simulation for further information.

types, cell pins and their capacitances. The resistance and capacitance values are in standard SPICE syntax and the cell instantiations are also included in this representation. The advantage of this format is that the DSPF file can be used as an input to a SPICE simulator itself. However, the drawback is that the DSPF syntax is too detailed and verbose with the result that the total file size for a typical block is very large. Thus, this format is rarely employed in practice for anything but a relatively small group of nets.

Here is an example DSPF file that describes the interconnect from a primary input *IN* to the input pin *A* of buffer *BUF*, and another net from output pin *OUT* of *BUF* to the primary output pin *OUT*.

```
.SUBCKT TEST_EXAMPLE OUT IN
* Net Section
*|GROUND_NET VSS
*|NET IN 4.9E-02PF
*|P (IN I 0.0 0.0 4.1)
*|I (BUF1:A BUF A I 0.0 0.7 4.3)
C1 IN VSS 2.3E-02PF
C2 BUF1:A VSS 2.6E-02PF
R1 IN BUF1:A 4.8E00
*|NET OUT 4.47E-02PF
*|S (OUT:1 8.3 0.7)
*|P (OUT O 0.0 8.3 0.0)
*|I (BUF1:OUT BUF1 OUT O 0.0 4.9 0.7)
C3 BUF1:OUT VSS 3.5E-02PF
C4 OUT:1 VSS 4.9E-03PF
C5 OUT VSS 4.8E-03PF
R2 BUF1:OUT OUT:1 12.1E00
R3 OUT:1 OUT 8.3E00
*Instance Section
X1 BUF1:A BUF1:OUT BUF
.ENDS
```

The nonstandard SPICE statements in DSPF are comments that start with “*|” and have the following format:

```
*|I(InstancePinName InstanceName PinName PinType PinCap X Y)
*|P(PinName PinType PinCap X Y)
```

```
*|NET NetName NetCap
*|S (SubNodeName X Y)
*|GROUND_NET NetName
```

4.3.2 Reduced Standard Parasitic Format

In the RSPF representation, the parasitics are represented in a reduced form. The reduced format involves a voltage source and a controlled current source. The RSPF format is also a SPICE file since it can be read into a SPICE-like simulator. The RSPF format requires that the detailed parasitics are reduced and mapped into the reduced format. This is thus a drawback of the RSPF representation since the focus of the parasitic extraction process is normally on the extraction accuracy and not on the reduction to a compact format like RSPF. One other limitation of the RSPF representation is that the bidirectional signal flow cannot be represented in this format.

Here is an example of an RSPF file. The original design and the equivalent representation are shown in Figure 4-13.

```
* Design Name : TEST1
* Date : 7 September 2002
* Time : 02:00:00
* Resistance Units : 1 ohms
* Capacitance Units : 1 pico farads
*| RSPF 1.0
*| DELIMITER " _ "
.SUBCKT TEST1 OUT IN
*| GROUND_NET VSS
*|NET CP 0.075PF
*|DRIVER CKBUF_Z CKBUF_Z
*|S (CKBUF_Z_OUTP 0.0 0.0)
R1 CKBUF_Z CKBUF_Z_OUTP 8.85
C1 CKBUF_Z_OUTP VSS 0.05PF
C2 CKBUF_Z VSS 0.025PF
*|LOAD Sdff1_CP Sdff1_CP
*|S (Sdff1_CP_INP 0.0 0.0)
E1 Sdff1_CP_INP VSS CKBUF_Z VSS 1.0
R2 Sdff1_CP_INP Sdff1_CP 52.0
C3 Sdff1_CP VSS 0.1PF
```

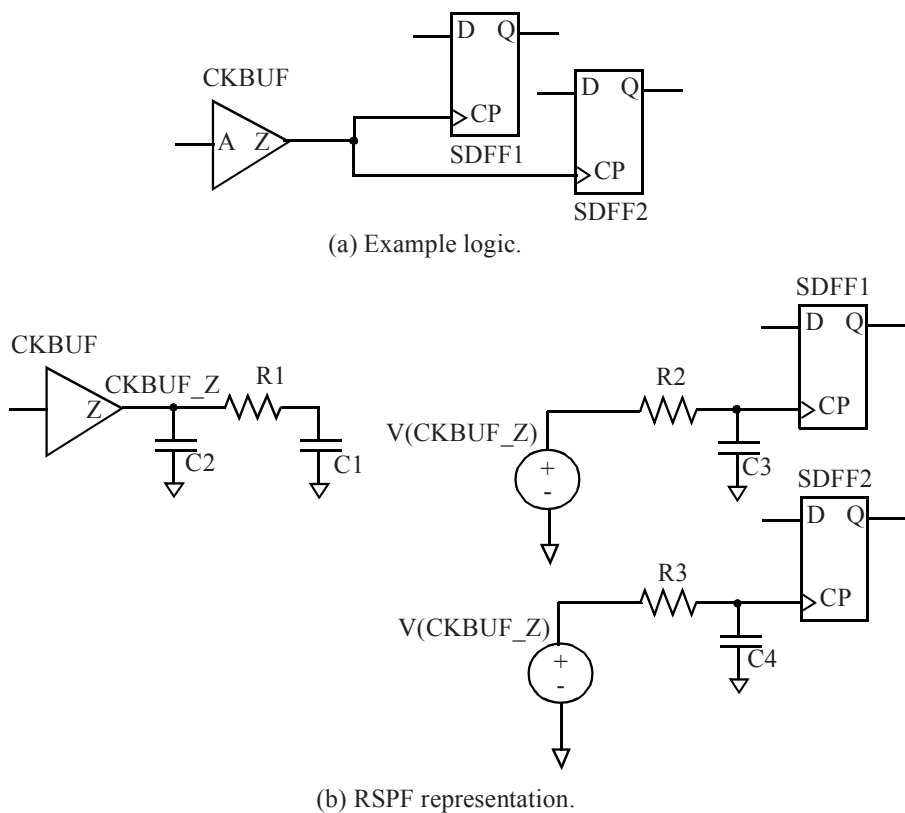


Figure 4-13 *RSPF example representation.*

```

*|LOAD SDFFF2_CP SDFFF2 CP
*|S (SDFFF2_CP_INP 0.0 0.0)
E2 SDFFF2_CP_INP VSS CKBUF_Z VSS 1.0
R3 SDFFF2_CP_INP SDFFF2_CP 43.5
C4 SDFFF2_CP VSS 0.1PF
*Instance Section
X1 SDFFF1_Q SDFFF1_QN SDFFF1_D SDFFF1_CP SDFFF1_CD VDD VSS SDFFF
X2 SDFFF2_Q SDFFF2_QN SDFFF2_D SDFFF2_CP SDFFF2_CD VDD VSS SDFFF
X3 CKBUF_Z CKBUF_A VDD VSS CKBUF
.ENDS
.END

```

This file has the following features:

- The pin-to-pin interconnect delays are modeled at each of the fanout cell inputs with a capacitor of value 0.1pF ($C3$ and $C4$ in the example) and a resistor ($R2$ and $R3$). The resistor value is chosen such that the RC delay corresponds to the pin-to-pin interconnect delay. The PI segment load at the output of the driving cell models the correct cell delay through the cell.
- The RC elements at the gate inputs are driven by ideal voltage sources ($E1$ and $E2$) that are equal to the voltage at the output of the driving gate.

4.3.3 Standard Parasitic Exchange Format

The SPEF is a compact format which allows the representation of the detailed parasitics. An example of a net with two fanouts is shown below.

```
*D_NET NET_27 0.77181
*CONN
*I *8:Q O *L 0 *D CELL1
*I *10:I I *L 12.3
*CAP
1 *9:0 0.00372945
2 *9:1 0.0206066
3 *9:2 0.035503
4 *9:3 0.0186259
5 *9:4 0.0117878
6 *9:5 0.0189788
7 *9:6 0.0194256
8 *9:7 0.0122347
9 *9:8 0.00972101
10 *9:9 0.298681
11 *9:10 0.305738
12 *9:11 0.0167775
*RES
1 *9:0 *9:1 0.0327394
2 *9:1 *9:2 0.116926
3 *9:2 *9:3 0.119265
4 *9:4 *9:5 0.0122066
```

```
5 *9:5 *9:6 0.0122066
6 *9:6 *9:7 0.0122066
7 *9:8 *9:9 0.142205
8 *9:9 *9:10 3.85904
9 *9:10 *9:11 0.142205
10 *9:12 *9:2 1.33151
11 *9:13 *9:6 1.33151
12 *9:1 *9:9 1.33151
13 *9:5 *9:10 1.33151
14 *9:12 *8:Q 0
15 *9:13 *10:I 0
*END
```

The units of the parasitics R and C are specified at the beginning of the SPEF file. A more detailed description of the SPEF is provided in Appendix C. Due to its compactness and completeness of representation, SPEF is the format of choice for representing the parasitics in a design.

4.4 Representing Coupling Capacitances

The previous section illustrated the cases where the capacitances of the net are represented as grounded capacitances. Since most of the capacitances in the nanometer technologies are sidewall capacitances, the proper representation for these capacitances is the signal to signal coupling capacitance.

The representation of coupling capacitances in DSPF is as an add-on to the original DSPF standard and is thus not unique. The coupling capacitances are replicated between both sets of coupled nets. This implies that the DSPF cannot be directly read into SPICE because of the duplication of the coupled capacitance in both set of nets. Some tools which output DSPF resolved this discrepancy by including half of the coupling capacitance in both of the coupled nets.

The RSPF is a reduced representation and thus not amenable to representing coupling capacitances.

The SPEF standard handles the coupling capacitances in a uniform and unambiguous manner and is thus the extraction format of choice when cross-talk timing is of interest. Further, the SPEF is a compact representation in terms of file size and is used for representing parasitics with and without coupling.

As described in Appendix C, one of the mechanisms to manage the file size is by having a name directory in the beginning of the file. Many extraction tools now specify a directory of net names (mapping net names to indices) in the beginning of the SPEF file so that the verbosity of repeating the net names is avoided. This reduces the file size significantly. An example with the name directory is shown in the appendix on SPEF.

4.5 Hierarchical Methodology

The large and complex designs generally require hierarchical methodology during the physical design process for the parasitic extraction and timing verification. In such scenarios, the parasitics of a block are extracted at the block level and can then be utilized at the higher levels of hierarchy.

The layout extracted parasitics of a block may be utilized for timing verification with another block whose layout has not been completed. In this scenario, layout extracted parasitics of layout-complete blocks are often used with wireload model based parasitic estimates for the pre-layout blocks.

In the case of the hierarchical flow, where the top level layout is complete but the blocks are still represented as black boxes (pre-layout), wireload model based parasitic estimates can be used for the lower level blocks along with the layout extracted parasitics for the top level. Once the layout of the blocks is complete, layout extracted parasitics for the top and the blocks can be stitched together.

Block Replicated in Layout

If a design block is replicated multiple times in layout, the parasitic extraction for one instantiation can be utilized for all instantiations. This requires that the layout of the block be identical in all respects for various instantiations of the block. For example, there should be no difference in terms of layout environment as seen from the nets routed within the block. This implies that the block level nets are not capacitively coupled with any nets outside the block. One way this can be achieved is by ensuring that no top level nets are routed over the blocks and there is adequate shielding or spacing for the nets routed near the boundary of the block.

4.6 Reducing Parasitics for Critical Nets

This section provides a brief outline of the common techniques to manage the impact of parasitics for critical nets.

Reducing Interconnect Resistance

For critical nets, it is important to maintain low slew values (or fast transition times), which implies that the interconnect resistance should be reduced. Typically, there are two ways of achieving low resistance:

- *Wide trace*: Having a trace wider than the minimum width reduces interconnect resistance without causing a significant increase in the parasitic capacitance. Thus, the overall RC interconnect delay and the transition times are reduced.
- *Routing in upper (thicker) metals*: The upper metal layer(s) normally have low resistivity which can be utilized for routing the critical signals. The low interconnect resistance reduces the interconnect delay as well as the transition times at the destination pins.

Increasing Wire Spacing

Increasing the spacing between traces reduces the amount of coupling (and total) capacitance of the net. Large coupling capacitance increases the crosstalk whose avoidance is an important consideration for nets routed in adjacent traces over a long distance.

Parasitics for Correlated Nets

In many cases, a group of nets have to be matched in terms of timing. An example is the data signals within a byte lane of a high speed DDR interface. Since it is important that all signals within a byte lane see identical parasitics, the signals are all routed in the same metal layer. For example, while metal layers *M2* and *M3* have the same average and the same statistical variations, the variations are independent so that the parasitic variations in these two metal layers do not track each other. Thus, if it is important for timing to match for critical signals, the routing must be identical in each metal layer.

□